

CREATYV

Social Autopilot

Developer-Ready Product & Technical Specification

MVP architecture • Desktop and mobile UX • AI system • Direct platform APIs • WhatsApp approvals

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Build decision

Build Creatyv as a responsive web application/PWA and a modular monolith. Use direct official APIs for publishing and WhatsApp messaging. Keep PostgreSQL as the source of truth, use queues for every external side effect, and make deterministic templates—not raw image generation—the default creative engine.

Document status	Decision
Audience	Founder, product designer, frontend engineer, backend engineer, AI engineer, QA and DevOps
Scope	Commercial MVP plus architecture that can scale to agency and multi-brand use
Primary launch integrations	Instagram Business, Facebook Pages, WhatsApp Cloud API and Google Business Profile
Phase 2 integration	LinkedIn organization publishing after access approval
Out of scope for MVP	Native apps, Canva-style editor, advanced video generation, social listening, ads and full agency white-labeling

00

How to use this specification

A delivery contract for product, design and engineering—not a conceptual feature list.

This document defines what is built, how it behaves, which systems own each state, and what “done” means. The development team should turn each page specification, domain command, integration flow and acceptance criterion into epics and tickets. Any third-party API assumption must be re-verified against the latest official documentation during implementation and before every platform-version upgrade.

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01

Product decisions and MVP boundaries

The MVP must make marketing happen reliably; AI generation is a supporting capability.

1.1 Product definition

Creatyv is an AI-assisted social media operations system for Indian businesses. A client configures the brand once, connects social accounts, chooses relevant festivals and approval rules, and then receives an automatically generated calendar, branded creatives, WhatsApp approval requests, scheduled publishing and confirmation reports.

Primary outcome

The key product metric is the number of brands that successfully publish approved content through Creatyv every month—not the number of AI generations.

1.2 MVP promise

Connect brand → Generate calendar → Create copy + template creative
→ Notify on WhatsApp → Approve / request changes
→ Schedule → Publish through official APIs → Confirm + report

1.3 Must-have launch scope

Capability	MVP decision	Definition of done
Brand setup	Build	Business basics, audience, language, voice, objectives, assets and restricted claims are persisted.
Calendar generation	Build	A week or month of valid slots is generated from structured brand rules and a strict JSON schema.
Creatives	Build	Template-based static creatives render consistently in 1:1 and 4:5; optional 9:16 is phase 1.1.
Approvals	Build	Individual and batch approval work in app; WhatsApp button approval creates a guarded domain transition.
Publishing	Build	Instagram, Facebook and GBP publish directly through official APIs with per-platform status.
Notifications	Build	WhatsApp, in-app and email fallback for approval, reminders, publish success/failure and reconnection.
Reports	Basic	Monthly publishing activity plus available reach/engagement metrics.
Billing	Build	Plan, limits, subscription, invoices and upgrade path are clear.

1.4 Explicit non-goals

- No full Canva-style editor. Customers can change text, image, template variant and crop—not freely position every element.
- No promise that AI generates production-quality brand imagery every time. Uploaded brand assets and deterministic templates are the default.
- No auto-publishing of unapproved content unless the owner has explicitly enabled Autopilot and accepted its rules.
- No automatic invention of prices, discounts, dates, medical claims, financial claims or legal statements.
- No native iOS/Android app for MVP. Build a fast responsive PWA with deep links from WhatsApp.
- No Zapier/Make dependency in the production architecture. They may be used only for internal prototypes, never as the system of record.

1.5 Product principles

Principle	Engineering implication
Outcome-first	Home shows approvals, upcoming posts, failures and connection health before generation tools.
Trust before autonomy	Preview, audit trail and approval are default; Autopilot is an explicit opt-in.
Structured AI	AI emits schema-valid data. Business logic, scheduling, permissions and status transitions remain deterministic.
Provider isolation	Every external platform is behind a provider adapter so API changes do not leak across the product.
Mobile approval-first	Mobile UX is designed around cards, preview, sticky approve/change actions and reminders.
Graceful failure	A failed provider call creates a visible status, retry path and ops event; it never disappears.

02

Users, roles and permissions

Keep customer roles simple while preserving workspace and tenant boundaries.

2.1 Tenant model

The top-level tenant is a Workspace. A Workspace can own one or more Brands. Each Brand owns assets, connected channels, content plans, posts and reports. This structure supports single-business customers now and agency/multi-brand use later without a migration.

Entity	Meaning	MVP rule
Workspace	Billing and membership boundary	One workspace is created on signup.
Brand	A business identity and content context	Starter/Autopilot defaults to one brand; higher plans can add brands.
Workspace member	A user with role-based access	Every request must be authorized against workspace membership.
Social account	A connected destination such as a Page or IG account	Belongs to one brand but credentials are stored under a secured connection.

2.2 MVP roles

Role	Can do	Cannot do
Owner / Admin	Manage brand, connections, approval rules, billing, members, publishing and all content.	Cannot bypass provider rules or modify audit history.
Editor	Create/edit plans, posts and assets; submit for approval.	Cannot connect accounts, enable Autopilot, change billing or approve own restricted post if policy forbids it.
Approver	Approve, request changes, skip and reschedule within allowed window.	Cannot change brand-level rules or billing.
Viewer	View calendar, published posts and reports.	No mutation actions.
Internal Ops	Support-only access through audited impersonation or admin tools.	No silent customer-data modification; every action logged.

2.3 Permission model

- **Authorization rule:** every API route resolves user → workspace membership → brand access → action permission.
- **Tenant isolation:** workspace_id is present on every tenant-owned table and every query is scoped server-side.
- **Sensitive actions:** connecting channels, enabling Autopilot, deleting data and changing payment require recent authentication.
- **WhatsApp approval:** the phone number must be mapped to an active approver and the one-time approval request must be unexpired.

03

Recommended technology stack

A TypeScript-first stack reduces handoff cost and keeps product, API and workers aligned.

3.1 Reference stack

Layer	Recommendation	Why
Frontend	Next.js + React + TypeScript	Responsive app, server rendering where useful, route-level loading/error states and a straightforward PWA path.
UI system	Tailwind CSS + Radix/shadcn primitives + design tokens	Fast consistent implementation with accessible dialogs, menus, tabs and form controls.
Forms/data	React Hook Form + Zod + TanStack Query	Typed validation, cached server state and predictable mutations.
Backend	NestJS + TypeScript, modular monolith	Clear modules, guards, dependency injection, OpenAPI and worker reuse without premature microservices.
ORM/database	PostgreSQL + Prisma + pgvector	Transactional source of truth, JSONB for evolving rules and vectors for Brand Brain retrieval.
Queue/cache	Redis + BullMQ	Delayed jobs, retries, rate limiting, locks and independent workers for rendering/publishing/notifications.
Object storage	Amazon S3 + CloudFront or Cloudflare R2 + CDN	Durable media storage and time-limited public URLs required by social publishing APIs.
Template renderer	SVG/HTML scene graph + Satori or custom SVG + Resvg + Sharp	Deterministic text/layout, image composition, resizing and export without building a design editor.
Identity	Managed auth provider or hardened custom auth	Use Google login, email OTP and optional phone OTP; avoid storing passwords unless necessary.
Payments	Razorpay Subscriptions for India; Stripe alternative	India-first recurring billing, invoices and webhook-driven entitlements.
Observability	Sentry + OpenTelemetry + structured logs; PostHog for product analytics	Correlate browser, API, queue and provider failures and measure funnel/retention.
Infrastructure	AWS ECS/Fargate + RDS + ElastiCache + S3; Vercel optional for frontend	Managed scaling without Kubernetes overhead in MVP.

Do not start with microservices

Separate deployable workers are useful; separate business services are not. Keep a modular monolith until team size, deployment frequency or scaling characteristics provide evidence for extraction.

3.2 Repository layout

```
apps/  
web/      # Next.js responsive client  
api/      # NestJS HTTP API + OAuth callbacks + webhooks  
worker-render/  # template and asset rendering jobs  
worker-publish/  # provider publishing and metrics jobs  
worker-notify/  # WhatsApp/email/in-app notifications  
packages/  
domain/    # entities, commands, state transitions, policies  
database/  # Prisma schema, migrations, repositories  
providers/ # Meta, WhatsApp, GBP, LinkedIn adapters  
ai/        # schemas, prompt versions, model router, evaluations
```

```
templates/      # template schema, renderer and validation
ui/             # shared tokens and components
observability/  # logging, tracing, error normalization
```

3.3 Environment strategy

Environment	Purpose	Key rules
Local	Developer integration with provider sandboxes/test assets where available	Use mocked provider adapters by default; never copy production tokens.
Development	Shared integration environment	Dedicated Meta/Google test app and test workspaces; synthetic WhatsApp contacts.
Staging	Release candidate and app-review environment	Production-like queues, CDN and callback domains; masked data only.
Production	Customer workloads	Separate cloud account/project, KMS keys, least privilege and protected migrations.

System architecture and backend modules

The database owns truth; queues execute side effects; providers are replaceable boundaries.



Creatyv MVP reference architecture

Modular monolith first; durable queues and provider adapters at the boundaries.

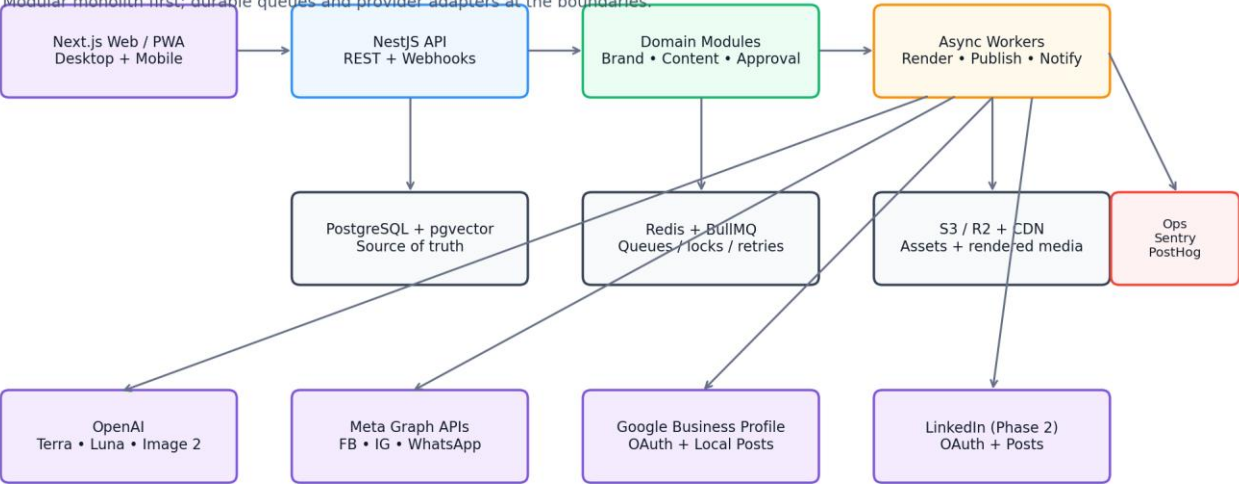


Figure 1. Reference architecture for the commercial MVP.

4.1 Backend modules

Module	Responsibility
Identity & Access	Users, sessions, memberships, RBAC, audit context.
Workspace & Billing	Plans, entitlements, usage limits, subscriptions and invoices.
Brand	Brand profile, voice, audiences, objectives, restrictions and festival preferences.
Assets	Upload, scanning, metadata, crop/focal point, tags, versions and signed URLs.
Connections	OAuth state, encrypted tokens, destinations, scope health and reconnect flow.
Content Planning	Deterministic slots, AI plan generation, validation and plan versions.
Posts	Post, per-platform variants, creative versions, comments and state transitions.
Approvals	Approval requests, batch approval, WhatsApp tokens and audit events.
Scheduling	Schedules, outbox events, due-job promotion and preflight checks.
Publishing	Provider adapters, attempts, errors, retries, permalinks and partial success.
Notifications	Template selection, preference/quiet-hour policy, WhatsApp/email/in-app delivery.
Festivals	Curated calendar, geography/relevance rules and brand campaign decisions.
Analytics	Metric snapshots, normalization and monthly reports.
AI Orchestration	Model routing, structured schemas, prompt versions, moderation and eval telemetry.

Module	Responsibility
Admin & Ops	Connection diagnostics, webhook replay, queue inspection, prompt/template release controls.

4.2 Command and event pattern

All mutations should be expressed as domain commands such as `ApprovePost`, `RequestChanges`, `SchedulePost` or `EnableAutopilot`. Commands validate the current state, authorization, version and business policy inside a transaction. External effects are emitted through a transactional outbox and processed asynchronously.

```
HTTP / webhook request
→ authenticate and authorize
→ execute domain command in PostgreSQL transaction
→ update entity + append audit record + append outbox event
→ commit
→ outbox dispatcher creates BullMQ job
→ worker calls provider adapter
→ result is persisted as PublishAttempt / NotificationDelivery
```

Critical consistency rule

Never update a post to `APPROVED` and enqueue a publish job in two unrelated operations. Use a transactional outbox so either both are committed or neither is.

05

Data model and state machines

Design the data model around versions, per-platform variants and immutable audit history.

5.1 Core entities

Entity	Minimum responsibilities / fields
User	id, email, phone, auth_provider_id, status, locale, timezone
Workspace	id, name, owner_user_id, plan_id, billing_status, default_timezone
WorkspaceMember	workspace_id, user_id, role, status, invited_at
Brand	id, workspace_id, name, industry, website, description, timezone, languages, status
BrandVoice	brand_id, tone, vocabulary, CTA rules, examples, prohibited phrases, version
BrandRule	brand_id, type, payload_json, severity, active
Asset	id, brand_id, type, storage_key, mime, dimensions, focal_point, tags, rights_status, status
SocialConnection	id, brand_id, provider, encrypted_credentials, scopes, expires_at, health_status
SocialAccount	id, connection_id, external_id, account_type, display_name, metadata_json, enabled
ContentPlan	id, brand_id, period_start, period_end, status, version, generation_run_id
ContentSlot	id, plan_id, local_date, local_time, pillar, objective, format, festival_id, priority
Post	id, brand_id, slot_id, title, status, version, approval_mode, source
PostVariant	id, post_id, social_account_id, platform, caption, hashtags, CTA, scheduled_at, status
Creative	id, post_id, template_version_id, content_json, rendered_asset_id, version, qa_status
Template / TemplateVersion	industry, format, aspect_ratio, schema_json, renderer_version, release_status
ApprovalRequest	id, post_id/batch_id, token_hash, approver_user_id, expires_at, status, post_version
ApprovalEvent	approval_request_id, actor, action, channel, metadata, created_at
Schedule	id, post_variant_id, publish_at_utc, timezone, status, version
PublishJob / PublishAttempt	provider, idempotency_key, attempt_no, status, response/error, timestamps
Notification	type, channel, recipient, template, status, related_entity, scheduled_for
WebhookEvent	provider, external_event_id, signature_valid, payload_ref, status, processed_at
Festival	name, date_rule, observed_date, regions, languages, categories, sensitivity
BrandFestivalPreference	brand_id, festival_id, mode, lead_days, campaign_type
PromptVersion / ModelRun	task, schema_version, model, prompt_hash, input/output refs, usage, latency, result
AuditLog	workspace_id, actor, action, entity_type/id, before/after summary, request_id, timestamp

5.2 Post state machine

Post lifecycle and guarded transitions

UI actions call domain commands; clients never set status fields directly.

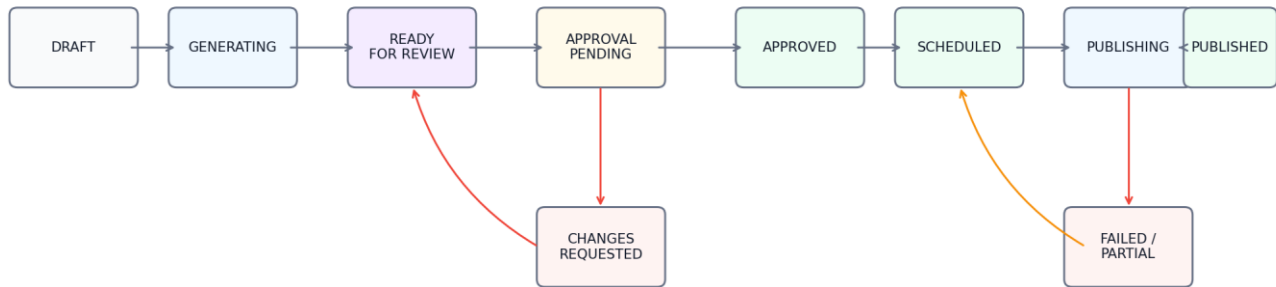


Figure 2. Post state machine. Per-platform delivery state is tracked separately.

Transition	Guard	Side effect
DRAFT → GENERATING	User can edit; generation quota available	Create ModelRun and generation job.
GENERATING → READY_FOR_REVIEW	Schema, claims and render QA pass	Create new post/creative version.
READY_FOR_REVIEW → APPROVAL_PENDING	Required assets and platform variants exist	Create approval request and notification outbox event.
APPROVAL_PENDING → APPROVED	Approver authorized; request valid; post version matches	Invalidate sibling requests and create scheduling events.
APPROVAL_PENDING → CHANGES_REQUESTED	Feedback supplied or structured reason selected	Create feedback thread and editing task.
APPROVED → SCHEDULED	Future time, connection health and entitlement valid	Persist schedules and enqueue preflight/due jobs.
SCHEDULED → PUBLISHING	Due, lock acquired, preflight valid	Create publish attempt.
PUBLISHING → PUBLISHED / PARTIAL / FAILED	Provider results persisted	Send confirmation or failure notification; schedule metrics.

5.3 Versioning rules

- Every material edit increments Post.version or Creative.version.
- An approval request records the exact post version. Any edit after approval invalidates that approval unless the edit is explicitly non-material.
- Published records are immutable. A correction is a new post/version, not an in-place history rewrite.
- Provider responses and webhook payloads are retained by reference with sensitive fields redacted.
- Prompt and template versions are pinned to generated output so results can be reproduced and compared.

06

Information architecture and navigation

Keep the primary navigation centered on work, not settings or AI tools.

6.1 Desktop navigation

Position	Item	Purpose
Workspace header	Brand switcher	Switch brand; display connection health and plan badge.
Primary sidebar	Home	Action center, upcoming posts, Autopilot health and monthly progress.
Primary sidebar	Calendar	Plan, preview and reschedule content.
Primary sidebar	Approvals	Individual/batch review and feedback.
Primary sidebar	Create	One-off post or new content plan.
Primary sidebar	Library	Assets, templates and previous content.
Secondary	Reports	Publishing and performance summaries.
Secondary	Brand	Profile, Brand Brain, assets, festivals and rules.
Secondary	Connections	Instagram, Facebook, GBP, WhatsApp and LinkedIn status.
Footer	Settings / Billing / Help	Member, notification, subscription and support controls.

6.2 Mobile navigation

Bottom tab	Default mobile view
Home	Action-required cards, today's posts, next festival and connection alerts.
Calendar	Agenda/list view by default; month picker as a filter, not a compressed grid.
Approvals	Full-screen swipe/card review with sticky Approve / Changes actions.
Create	Bottom sheet: one-off post, generate week, generate month, upload asset.
More	Library, reports, brand, connections, billing, settings and support.

6.3 Global interaction patterns

- **Global status bar:** show “Autopilot active”, “Approval required” or “Connection needs attention” with one clear action.
- **Platform badges:** every post card shows exact destinations and each platform's state.
- **Sticky mobile actions:** Approve, Request changes and Edit remain reachable while scrolling a preview.
- **Drawers over navigation:** desktop post details open in a right drawer for quick review; full page for deep editing.
- **Plain language:** use “Reconnect Instagram” rather than “OAuth token expired”.
- **No hidden failures:** failed posts appear on Home, Calendar and Notifications until resolved.

07

Signup and onboarding pages

Progressively collect enough brand context to create value without a long empty setup.

Page	Desktop / mobile UX	Primary inputs/actions	Backend	Critical states
7.1 Signup / login	Desktop: centered authentication card with Creatyv outcome copy and trust statement. Mobile: full-width form, large tap targets and no marketing carousel.	Google sign-in, email magic link/OTP, optional phone OTP, terms consent.	POST /auth/*; callback creates User and empty Workspace.	Loading, expired OTP, account collision and suspended-account states.
7.2 Workspace and brand creation	Two-column desktop: form left, "what happens next" summary right. Mobile: one question per screen.	Workspace name, brand name, website, industry, timezone and preferred language.	POST /workspaces; POST /brands.	Offer website import but never block manual setup.
7.3 Business basics	Stepper with progress and autosave. Show examples based on industry.	Description, products/services, locations, customer type and primary objective.	PATCH /brands/:id; POST /brand-import jobs.	Extracted website data appears as editable suggestions.
7.4 Audience and voice	Desktop: selectable cards plus examples; mobile: chips and short text fields.	Audience, age/segment, tone, preferred words, avoided words, languages and sample captions.	PUT /brands/:id/voice; POST /brands/:id/voice/suggest .	Show generated tone sample and "sounds like us / adjust" feedback.
7.5 Brand assets	Grid uploader with asset quality checks. Mobile camera/gallery upload and crop.	Logo variants, colors, fonts, product/service images, founder/team, menu/catalogue.	POST /assets/upload-url; POST /assets; PATCH /assets/:id.	Flag low resolution, missing usage rights and unsupported formats.
7.6 Content preferences	Simple business language; show suggested mix.	Posts/week, content topics, platforms, days/times, campaign vs evergreen ratio.	PUT /brands/:id/content-preferences.	Prevent plan selections beyond subscription entitlement.
7.7 Festivals and occasions	Region-aware searchable calendar; default recommendations unchecked.	National, regional, religious, industry and custom dates; greeting vs promotion; lead time.	GET /festivals; PUT /brands/:id/festival-preferences.	Never enable every festival automatically.
7.8 Connect social accounts	Connection cards with benefit, permission explanation and status.	Connect Meta and GBP; select Page, IG account and GBP location.	GET /connections/:provider/oauth/start; OAuth callbacks; GET /connections/status.	Show missing professional account/admin permission and reconnect steps.
7.9 WhatsApp opt-in	Explain notification categories and quiet hours. Send a verification message.	Phone, consent, language, quiet hours and reminder frequency.	POST /whatsapp/opt-in; POST /whatsapp/verify; webhook confirms.	Do not silently reuse a phone number from billing.
7.10 Approval mode	Risk-aware cards; "Approve every post" default.	Every post, weekly batch, regular auto-publish + festival approval, full Autopilot.	PUT /brands/:id/approval-policy.	Enabling Autopilot requires explicit acknowledgment and recent authentication.
7.11 First calendar generation	Show progress stages and early preview; do not leave a blank dashboard.	Generate 7-day starter plan first; then offer month generation.	POST /content-plans/generate; GET /jobs/:id.	On failure, keep collected setup and offer retry/support.
7.12 First approval and completion	Guided post preview with highlighted approve/change controls.	Approve first post, choose schedule and see confirmation.	POST /approvals/:id/approve; POST /schedules.	Completion screen shows next post and WhatsApp test notification.

7.13 Onboarding completion logic

Requirement	Mandatory before generation	Mandatory before publishing
Brand name, industry, timezone and description	Yes	Yes

Requirement	Mandatory before generation	Mandatory before publishing
At least one objective and content category	Yes	Yes
Brand voice and restricted-claim confirmation	Yes	Yes
Logo and two usable images	Recommended; template can use text-only fallback	Required for image-led template posts
Connected platform	No; preview can be generated	Yes, for that destination
WhatsApp opt-in	No	No, but required for WhatsApp approvals
Approval policy	Default can be created	Yes
Billing entitlement	Trial can generate sample	Active entitlement required for live publishing

08

Core application pages: desktop and mobile

Each page must have one dominant job and visible state ownership.

Page	Desktop / mobile structure	Key actions	Backend/read model	States to design
8.1 Home / Action Center	Desktop: KPI strip, Action Required column, Upcoming Content timeline, Autopilot health and recent activity. Mobile: action cards first, then today/this week.	Approve posts, resolve failed publish, reconnect account, answer festival prompt, create plan.	GET /dashboard; aggregated read model.	Skeleton loading, zero-action success state, setup incomplete, API incident banner.
8.2 Calendar	Desktop month/week/list with filters, drag reschedule and right-side preview. Mobile agenda by day with month/date picker.	Open post, reschedule, filter by platform/status, generate empty slots, batch select.	GET /calendar; PATCH /posts/:id/schedule; POST /calendar/batch-actions.	Conflict, outside plan limits, expired approval and partial-platform status.
8.3 Approvals	Desktop split view: queue left, post preview right; batch toolbar. Mobile one card at a time with sticky actions.	Approve, request changes, skip, edit, batch approve, change schedule.	GET /approvals; POST /approvals/:id/{approve,changes,skip}.	Version mismatch, expired request, unauthorized approver, already actioned.
8.4 Create	Choice-based landing: Create one post, Generate week/month, Create festival campaign, Reuse content.	Start guided workflow; never expose raw prompt box as primary UI.	POST /posts; POST /content-plans/generate.	Quota reached, insufficient assets, no connection.
8.5 Plan generator	Wizard: period, objective, frequency, platforms, topics, campaigns, exclusions; review slot skeleton before full generation.	Generate plan, adjust slots, regenerate selected slots only.	POST /content-plans/generate; PATCH /content-plans/:id/slots.	Model timeout, duplicate topics, unavailable dates, plan version conflict.
8.6 Post detail / editor	Desktop 3-panel: creative preview, content controls, platform preview/status. Mobile tabs: Preview / Caption / Schedule; sticky submit.	Edit caption, regenerate selected field, replace asset, template variant, crop, schedule, submit approval.	GET/PATCH /posts/:id; POST /posts/:id/regenerate; POST /creatives/:id/render.	Unsaved edits, approval invalidation, render overflow, provider requirement error.
8.7 Creative template editor	Customer-facing controls only: text fields, image, focal point, brand color variant, template choice.	Preview 1:1 and 4:5, switch template, download preview if entitlement allows.	PATCH /creatives/:id/content; POST /creatives/:id/render.	Text overflow, low contrast, low-resolution image and missing logo.
8.8 Library	Tabs: Assets, Templates, Drafts, Published, Saved Ideas. Mobile search/filter plus upload FAB.	Upload, tag, archive, reuse, rights mark, set primary logo/focal point.	GET/POST /assets; GET /posts?status=; POST /posts/:id/duplicate.	Unsupported file, duplicate, processing, rights unknown.
8.9 Festivals / campaigns	Timeline of selected occasions, decision cards and campaign status. Mobile upcoming list.	Enable/disable, greeting/campaign/skip, add offer details, generate campaign.	GET /festivals/recommended; PUT preferences; POST /campaigns.	Date changes, regional mismatch, sensitive occasion warning.
8.10 Connections	Provider cards with destinations, scopes, health, last successful publish and reconnect.	Connect, select destinations, test, refresh/reconnect, disconnect.	/connections/* and provider callbacks.	Expired token, missing scope, account role changed, provider outage.
8.11 Brand / Brand Brain	Sections: Basics, Voice, Audience, Rules, Assets, Content preferences, Learning history.	Edit structured context, add prohibited claims, approve examples, reset learned preference.	GET/PATCH /brands/:id/*; POST /brand-brain/reindex.	Unsaved changes, conflicting rules, reindexing.
8.12 Notifications	Inbox grouped by required action, publishing, account health and reports.	Mark read, open related object, notification preferences.	GET /notifications; PATCH read; PUT preferences.	Delivery failure and channel disabled.
8.13 Reports	Desktop monthly summary with publish completion, reach/engagement where available and content insights. Mobile card summary.	Change period, export PDF/CSV later, open top post, view recommendations.	GET /reports/monthly; GET /analytics/*	Metrics delayed/unavailable and platform-specific definitions.
8.14 Team	Member list, roles and pending invites.	Invite, change role, revoke; designate WhatsApp approvers.	POST /members/invite; PATCH /members/:id; DELETE.	Invite expired, last owner protection.

Page	Desktop / mobile structure	Key actions	Backend/read model	States to design
8.15 Billing	Current plan, usage meter, payment method, invoices and upgrade comparison.	Upgrade/downgrade, cancel renewal, add brand/posts.	/billing/*; Razorpay webhooks.	Payment pending/failed, entitlement grace period.
8.16 Settings / support	Personal profile, security, timezone, notification quiet hours, data/export/delete, support tickets.	Update preferences, sign out devices, export data, contact support.	/me, /sessions, /privacy/*, /support/tickets.	High-risk action confirmation and support SLA messaging.
8.17 Internal admin / ops	Customer-search, read-only diagnostics, queue failures, connection health, webhook replay, template/prompt/festival releases.	Replay safe webhook, retry job, revoke connection, publish new template version, feature flag.	/admin/* with separate authorization and audit.	No direct production DB editing; break-glass actions require reason.

8.18 Recommended dashboard hierarchy

20. Action Required: approvals, failed publishes, expired connections and missing campaign information.
21. Today and Next: scheduled posts with per-platform status and one-click preview.
22. Autopilot Health: active mode, last successful publish, next planned run and any blockers.
23. Upcoming Festivals: only selected/relevant occasions with campaign decision deadline.
24. Monthly Progress: planned, approved, scheduled, published and skipped—not vanity analytics.
25. Performance Snapshot: available reach/engagement and top post after the operational sections.

Mobile rule

Do not show a desktop month calendar scaled down. Default mobile Calendar to a chronological agenda with day chips, filters and post cards.

09

Content calendar generation engine

Use deterministic scheduling around AI planning; never ask a model to own dates and publishing rules.

9.1 Generation pipeline

26. Load brand profile, voice, objectives, restrictions, selected festivals, plan entitlement, connected platforms and timezone.
27. Build deterministic candidate slots in code from period, posts/week, preferred days/times, blackout dates and provider constraints.
28. Reserve mandatory or selected festival/campaign slots through a rule engine; never create all festivals automatically.
29. Retrieve recent posts and embedding matches to enforce topic, headline and visual diversity.
30. Ask the planning model for a structured plan skeleton mapped to the candidate slot IDs—not arbitrary dates.
31. Validate schema, slot ownership, duplicates, forbidden claims, offer fields and content mix. Reject or repair invalid output.
32. Generate platform-specific copy for each accepted slot with a high-volume model.
33. Select a compatible template and approved asset using deterministic rules and model-provided semantic fields.
34. Render creative variants and run automated QA for dimensions, overflow, safe areas, contrast, file size and required branding.
35. Create Post, PostVariant, Creative and ApprovalRequest records; send one grouped notification rather than one message per post.

9.2 Slot-builder rules

Rule	Implementation
Timezone	Store brand IANA timezone; convert to UTC only when creating schedules.
Frequency	Generate exact slot count based on plan and content preference.
Day/time	Use preferred windows; spread posts; avoid duplicate platform posts at identical times unless configured.
Blackouts	Respect business closures, user-excluded dates and platform connection downtime.
Festival lead time	Campaign slots can begin before festival; greeting slot lands on approved date.
Content diversity	Limit repeated pillar, CTA, template and opening hook across rolling period.
Platform fit	LinkedIn text, GBP update/offer and Instagram visual formats receive different variants.
Entitlements	Reject slots beyond posts/month, connected channels or brand limit before AI calls.

9.3 Structured plan schema

```
{
  "plan_version": "1.0",
  "period": {"start": "2026-08-01", "end": "2026-08-31"},
  "slots": [{
    "candidate_slot_id": "slot_...",
    "content_pillar": "product_education",
    "objective": "build_trust",
    "format": "static",
    "platform_targets": ["instagram", "facebook", "gbp"],
    "concept": "Explain the product benefit using approved facts",
    "creative_text": {"headline": "...", "subheadline": "..."},
    "asset_requirements": ["product_image"],
    "festival_id": null,
    "offer_reference_id": null,
    "approval_required": true,
  }]
```

```
"sensitivity": "normal"
  }
}
```

9.4 Validation and repair

Validation	Fail behaviour
Unknown slot ID or out-of-period date	Reject plan output and retry once with validation feedback.
Invented offer/price/date	Reject slot; require linked OfferReference or explicit client input.
Duplicate concept/headline	Replace only offending slot; keep accepted slots.
Restricted claim detected	Block and route to human review.
Template lacks required asset	Choose fallback template or mark "asset required".
AI call fails	Keep deterministic slot skeleton and allow retry; do not lose user selections.

Cost and UX rule

Generate a seven-day sample during onboarding. Generate a full month only after the user accepts the direction or starts a paid plan. Regenerate individual slots, not the whole calendar, wherever possible.

10

AI models, prompts and Brand Brain

Use model routing by task, strict schemas and observable prompt versions.

10.1 Recommended model routing (as of 17 July 2026)

Task	Recommended model	Reason / fallback
Monthly strategy and calendar skeleton	GPT-5.6 Terra	Strong reasoning/cost balance. Escalate rare complex campaign cases to GPT-5.6 Sol.
High-volume captions and platform adaptations	GPT-5.6 Luna	Lower-cost generation for structured repetitive tasks; retry with Terra when validation fails.
Tone rewrite, hashtag/CTA and notification summaries	GPT-5.6 Luna	Fast, cost-sensitive workload.
Complex internal QA or policy reconciliation	GPT-5.6 Terra / Sol	Use only when deterministic checks cannot resolve ambiguity.
Optional concept/background images	GPT Image 2	Use for backgrounds or concept assets; not for final logo/text placement.
Brand Brain retrieval	text-embedding-3-small	Cost-effective multilingual embeddings; upgrade selectively if retrieval quality demands it.
Safety screening	OpenAI moderation endpoint + deterministic claim rules	Model moderation complements—not replaces—industry-specific validators.

Model names change

Keep all model IDs in configuration and route through an AIProvider interface. No business code should hard-code a model. Re-run evaluations before changing the default model or prompt version.

10.2 Brand Brain design

The Brand Brain is not an autonomous agent. It is a retrieval and rules layer that assembles approved context for each generation task. Structured facts should be preferred over embeddings; vector retrieval should provide examples and supporting material.

Context tier	Examples	How used
Structured facts	Products, locations, audience, languages, approved offers, opening hours	Inserted directly into schemas and validators.
Hard rules	Prohibited claims, mandatory disclaimers, tone constraints, festival exclusions	Applied before and after generation; cannot be overridden by model.
Approved examples	Previously accepted captions, headlines and template choices	Retrieved by similarity and diversity; used as style references.
Rejected examples	Client feedback and rejection reason	Used as negative constraints, not copied into prompts verbatim without sanitization.
Source documents	Menu, catalogue, brochure and brand guide chunks	Retrieved with citations/internal source IDs to ground factual copy.

10.3 Prompt and model-run controls

- Use the OpenAI Responses API and strict Structured Outputs/JSON Schema for every machine-consumed response.

- Persist PromptVersion with task, schema version, instructions hash, model, temperature/reasoning configuration and release status.
- Persist ModelRun with token usage, cost estimate, latency, validation errors, repair count and resulting entity IDs.
- Never pass provider access tokens, payment data or unnecessary personal data to the model.
- Redact or summarize customer feedback before placing it into long-lived context.
- Use a golden evaluation set per vertical and measure schema success, factual error rate, approval rate, edit distance and cost per approved post.

10.4 AI output boundary

AI MAY produce:

concepts, semantic content fields, captions, hashtags, CTAs,
platform adaptations, suggested template type, asset requirements.

AI MUST NOT directly:

create schedules, connect accounts, approve content, publish content,
invent offers/prices, change billing, or mutate status fields.

11

Template-based creative generation

AI supplies meaning; the renderer owns typography, brand fidelity and platform dimensions.

11.1 Template architecture

Component	Specification
Template definition	Versioned JSON/SVG scene graph with slots for background, image, logo, headline, subheadline, CTA and optional offer.
Aspect ratios	Minimum 1:1 and 4:5 at launch; add 9:16 for stories/reels covers and 16:9 where required.
Content contract	Every template declares max characters, required asset types, safe area, alignment rules and color variants.
Renderer	Server-side deterministic SVG/HTML render to PNG/JPEG; use Sharp for resize/compression and metadata stripping.
Font handling	Only licensed bundled fonts or customer-uploaded fonts with rights declaration; preload and validate glyph coverage.
Brand colors	Store tokens and check contrast; template defines allowable role mapping, not arbitrary user placement.
Asset crop	Store focal point and crop per aspect ratio; render previews before approval.
QA	Overflow, missing glyph, logo size, contrast, image resolution, file size, dimensions and safe areas.

11.2 Creative generation flow

Post concept + brand rules + selected asset
 → AI returns semantic fields only
 → template matcher scores compatible templates
 → renderer composes SVG/HTML
 → image pipeline exports platform variants
 → automated QA
 → store immutable rendered asset version
 → show preview / request approval

11.3 Customer editing surface

Allow in MVP	Do not allow in MVP
Edit headline/subheadline/CTA within limits	Freeform drag and resize of every object
Switch among approved template variants	Arbitrary font installation during editing
Replace image and adjust focal point/crop	Layer panel and complex masks
Select brand-approved color variant	Unrestricted colors outside brand kit
Preview platform ratios	Pixel-perfect manual desktop layout on mobile
Regenerate copy fields independently	Regenerate the entire design when only one field changed

11.4 Optional generative imagery

Use GPT Image 2 only for optional background creation, controlled image editing, texture, illustration or concept assets. Final typography, logo and CTA must still be placed by the deterministic renderer. Store the generation prompt, source asset references and usage rights metadata with the generated asset.

Why this matters

A template pipeline gives predictable brand consistency, fast rendering, low cost and easy approvals. Raw AI imagery should be a feature—not the foundation of the monthly deliverable.

12

Direct social publishing integrations

Use official APIs behind a common provider adapter; start app reviews early.

12.1 Provider adapter contract

```
interface SocialPublisher {
  startOAuth(input): Promise<OAuthStart>
  completeOAuth(callback): Promise<ConnectionResult>
  listDestinations(connection): Promise<Destination[]>
  validateConnection(connection): Promise<ConnectionHealth>
  publish(request, idempotencyKey): Promise<PublishResult>
  getPublishStatus(externalId): Promise<PublishStatus>
  getPermalink(externalId): Promise<string | null>
  fetchMetrics(externalId, range): Promise<MetricSnapshot[]>
  revoke(connection): Promise<void>
}
```

12.2 Meta: Instagram Business and Facebook Pages

Stage	Implementation requirement
App setup	Create Meta app, business verification, privacy policy, data deletion URL, required products and app-review submissions.
OAuth	Use state and PKCE where supported; request only required scopes; map user-managed Pages and connected professional Instagram accounts.
Token storage	Encrypt tokens with KMS envelope encryption; store scopes, expiry and token owner; never return tokens to browser.
Instagram media	Host final media at a provider-fetchable signed/public URL; create a media container through IG User Media.
Video/Reel readiness	Poll container status before publish; store provider container ID.
Instagram publish	Publish prepared container via IG User Media Publish; record returned media ID and later permalink/metrics.
Publishing limit	Query the content publishing limit endpoint; do not hard-code a number because platform rules can change.
Facebook Page post	Use the current Pages API for Page feed/photo/video publishing; store Page post ID and per-destination response.
Webhooks	Subscribe to relevant account/status changes where available; also run scheduled connection health checks.

12.3 Instagram publish sequence

36. Worker validates connection, media URL, caption limits, asset dimensions and publishing entitlement.
37. Create the Instagram media container with image_url or video_url and the relevant media_type.
38. For asynchronous media, poll container status with bounded retries.
39. Check current publishing usage/limit where appropriate.
40. Call media_publish with the creation/container ID.
41. Persist the returned external media ID, response metadata and publish timestamp.
42. Fetch permalink and metrics asynchronously; never block publish confirmation on analytics.

12.4 Google Business Profile

Stage	Implementation requirement
Access	Register the application, obtain OAuth credentials and complete Google Business Profile API access requirements before launch.
OAuth scope	Use the current business management scope; select an account and location after authorization.
Post types	Support update/news, event and offer posts plus supported CTAs; do not claim API support for product posts.
Create endpoint	Create LocalPost under the selected account/location using the current v4 endpoint.
Validation	Use provider validation capabilities when available and enforce summary/media/CTA requirements internally.
Metrics	Fetch local post insights in a background job where available and normalize separately from Meta metrics.
Scheduling	Schedule inside Creatyv and call GBP at due time; do not depend on provider-side recurrence for MVP.

12.5 LinkedIn (Phase 2)

- Implement only after the product is accepted for the required LinkedIn Marketing/Community Management access.
- For organization publishing, verify the authenticated member's organization role and required scopes.
- Use the same provider adapter and per-platform variant model; do not block MVP launch on LinkedIn approval.
- Keep organization text/image/document publishing separate from personal member sharing permissions.

12.6 Connection health model

Health status	Meaning	User action
HEALTHY	Token/scopes valid and destination accessible	None
EXPIRING	Expiry or provider warning approaching	Reconnect before stated date
MISSING_SCOPE	Required permission absent	Reauthorize with explanation
ROLE_CHANGED	User no longer administers destination	Use another admin or reconnect
REVOKED	Provider/user revoked access	Connect again
PROVIDER_INCIDENT	External outage or widespread error	No client blame; status page and automatic retry
UNKNOWN	Health check not completed	Retry check; prevent unsafe publish

13

WhatsApp approval and reminder system

WhatsApp is the convenience layer; Creatyv remains the authenticated source of truth.

WhatsApp approval-to-schedule loop

A button press is an authenticated event, not a direct database update.

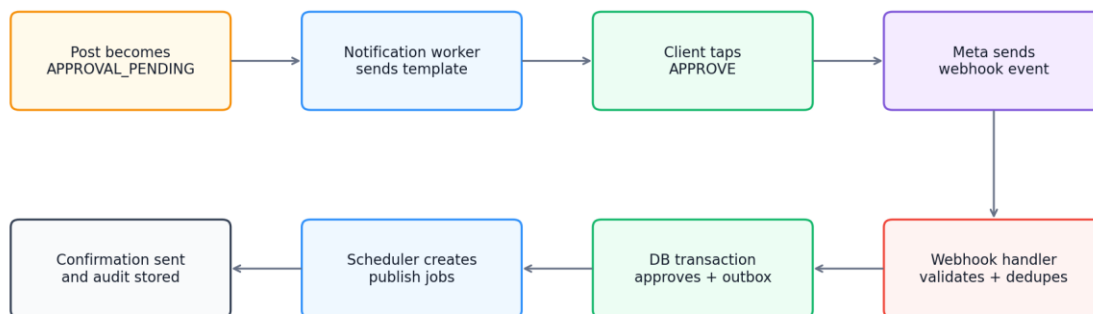


Figure 3. Secure WhatsApp approval-to-schedule workflow.

13.1 Integration choice

Use the WhatsApp Business Platform Cloud API directly through Meta. For MVP, messages can originate from a Creatyv business number after explicit user opt-in. If customers later require their own WhatsApp number, add Meta Embedded Signup as a separate phase and preserve the same WhatsAppProvider interface.

13.2 Message templates

Template key	Trigger	Buttons / CTA
post_ready	A post or batch enters APPROVAL_PENDING	Approve • Open preview • Request changes
approval_reminder	No action by configured reminder time	Approve • Open preview • Request changes
revision_ready	A requested revision is ready	Approve • Open preview • Request changes
scheduled_confirmation	Approval created schedules	View calendar • Change schedule
published_success	All selected platforms published	View post / View report
published_partial	Some platforms succeeded and some failed	View status • Retry failed
publish_failed	No destination published	View issue • Retry / Reconnect
connection_attention	Account scope/token/role problem	Reconnect account
festival_decision	Selected festival lead window reached	Create campaign • Greeting only • Skip
monthly_report_ready	Report generated	View report

13.3 Approval button security

43. Create an ApprovalRequest tied to a specific post version, approver, workspace, expiry and allowed actions.
44. Generate a high-entropy random token, store only its hash, and map it to a short opaque button reply ID. Do not expose raw database IDs.
45. Send the approved WhatsApp template. Persist Meta message ID and notification delivery record.

46. When the user taps Approve, Meta sends a messages webhook containing the button reply ID.
47. Validate webhook signature/challenge, deduplicate external event ID and resolve the button ID to the ApprovalRequest.
48. Confirm authorized phone, active membership, allowed action, request expiry, post state and post version.
49. Execute ApprovePost inside one DB transaction: set approved state, append ApprovalEvent/AuditLog and append scheduling outbox events.
50. Commit, enqueue schedules through the outbox dispatcher and send an acknowledgement message.

13.4 Webhook handler pseudocode

```

POST /webhooks/whatsapp
verify_meta_signature(rawBody)
event = parse_and_normalize(rawBody)
if WebhookEvent.exists(event.externalId): return 200

persist WebhookEvent(RECEIVED)
if event.type == 'interactive_button_reply':
    request = findApprovalRequestByOpaqueReplyId(event.replyId)
    assert request.phone == normalize(event.senderPhone)
    assert request.status == 'PENDING' and request.expiresAt > now
    assert request.postVersion == currentPost.version
    transaction:
        Approvals.approve(request, actor='WHATSAPP')
        Outbox.append('POST_APPROVED', {postId, version})
        Audit.append(...)
    mark WebhookEvent(PROCESSED)
return 200

```

13.5 Reminder policy

When	Rule	Fallback
Immediately when ready	Send one grouped approval request; obey quiet hours unless urgent.	In-app notification always.
First reminder	Default 24 hours before scheduled deadline or configurable interval.	Do not send if already actioned.
Final reminder	Default 2 hours before approval deadline.	Highlight that post will be held/skipped.
No response	Default HOLD; never publish without explicit Autopilot policy.	Move to unresolved queue and notify dashboard.
Autopilot enabled	Approval may be implicit for eligible regular content only.	Festival/sensitive/offer posts still follow policy.
Notification failure	Retry transient errors; email/in-app fallback.	Surface invalid/opted-out number.

WhatsApp limitation

Interactive reply buttons support up to three predefined replies. Use Approve, Open Preview and Request Changes; place Skip and advanced editing on the secure preview page if needed.

13.6 Deep-link approval page

The Open Preview button should use a short-lived signed deep link to a mobile-optimized post preview. If the user is already authenticated, open directly. Otherwise use OTP/magic-link verification and preserve the intended action. The preview page must still re-check current version and permission before approving.

14

Scheduler, retries and idempotency

Reliable publishing requires a durable schedule, preflight checks and recoverable failure classes.

14.1 Scheduling architecture

PostgreSQL Schedule row (source of truth)

- transactional outbox event
- BullMQ delayed preflight job (e.g. T-15 minutes)
- connection/media/policy validation
- BullMQ due publish job
- provider adapter call under distributed lock
- PublishAttempt persisted
- result event → confirmation / retry / dead-letter queue

14.2 Idempotency

- **Internal key:** postVariantId:provider:scheduledAt:postVersion.
- **Database protection:** unique constraint on active publish job idempotency key.
- **Worker lock:** Redis lock or PostgreSQL advisory lock around provider call.
- **Provider uncertainty:** if timeout occurs after request submission, query provider/status before re-sending where possible.
- **Webhook deduplication:** unique provider + external_event_id.
- **Notification deduplication:** notification type + related entity + recipient + policy version.

14.3 Error classification and retry policy

Class	Examples	Action
Validation / permanent	Unsupported media, caption violation, missing required field	Do not retry automatically; mark failed and return actionable message.
Authentication	Expired/revoked token, missing permission, role changed	Pause destination, request reconnect, retry only after connection healthy.
Rate limit	Provider quota or HTTP 429	Respect retry-after; exponential backoff with jitter; throttle provider queue.
Transient provider	5xx, network timeout, temporary processing error	Bounded exponential retries; preserve idempotency.
Media processing	Video/container still processing	Poll with maximum age; fail to ops if not ready.
Unknown response	Timeout after submission or ambiguous provider state	Reconcile before retry to avoid duplicate post.
Internal	Renderer crash, DB/queue incident	Retry safely; alert engineering; keep visible state.

14.4 Per-platform result model

A multi-platform post is successful per destination, not atomically across all destinations. The parent Post can become PUBLISHED when all succeed, PARTIALLY_PUBLISHED when at least one succeeds and one fails, or FAILED when none succeed. Users must be able to retry only the failed variants.

14.5 Time handling

- Collect and store an IANA timezone per brand and optional per location.
- Store publish_at_utc plus the original local time and timezone for audit/display.
- Handle daylight-saving changes for international customers even if the India launch is IST-centric.
- Use UTC in queues and database comparisons; render local date/time in the UI.

- Define a “too late” policy: approved after deadline either publish now, move to next slot or hold, based on brand policy.

15

Application API surface

Expose task-oriented endpoints; do not expose generic status mutation.

Domain	Representative endpoints	Purpose
Auth	POST /auth/otp/request; POST /auth/otp/verify; GET /auth/oauth/:provider; GET /auth/callback	Authentication and account linking.
Workspaces	GET/POST /workspaces; GET/PATCH /workspaces/:id; GET/POST/PATCH /members	Tenant and membership management.
Brands	GET/POST/PATCH /brands; GET/PUT /brands/:id/voice rules preferences	Structured Brand Brain inputs.
Assets	POST /assets/upload-url; POST/GET/PATCH/DELETE /assets	Direct upload with metadata and processing.
Connections	GET /connections; GET /connections/:provider/oauth/start; GET /oauth/:provider/callback; POST /connections/:id/test; DELETE	Provider OAuth and health.
Festivals	GET /festivals; GET /festivals/recommended; PUT /brands/:id/festival-preferences	Curated event preferences.
Plans	POST /content-plans/generate; GET /content-plans/:id; PATCH /content-plans/:id/slots; POST /slots/:id/regenerate	Plan generation and selective repair.
Posts	GET/POST /posts; GET/PATCH /posts/:id; POST /posts/:id/regenerate submit-review duplicate	Post editing and workflow commands.
Creatives	PATCH /creatives/:id/content; POST /creatives/:id/render; GET /creatives/:id/preview	Deterministic creative render.
Approvals	GET /approvals; POST /approvals/:id/approve request-changes skip; POST /approval-batches/:id/approve	Guarded approval actions.
Calendar	GET /calendar; POST /calendar/batch-actions; POST /posts/:id/reschedule	Calendar read and schedule commands.
Publishing	POST /post-variants/:id/retry; GET /publish-attempts/:id	Retry failed destination and inspect status.
Notifications	GET /notifications; PATCH /notifications/:id/read; PUT /notification-preferences	In-app notification center.
Reports	GET /reports/monthly; GET /analytics/posts/:id	Normalized activity and metrics.
Billing	GET /billing/subscription usage invoices; POST /billing/checkout change-plan cancel	Subscription and entitlements.
Webhooks	POST /webhooks/meta; /webhooks/whatsapp; /webhooks/google; /webhooks/razorpay	Raw-body verified provider events.
Admin	GET /admin/jobs connections webhooks; POST /admin/jobs/:id/retry; POST /admin/webhooks/:id/replay	Audited operational controls.

15.1 API conventions

- Generate OpenAPI from NestJS decorators and publish a versioned internal API contract.
- Use cursor pagination for posts, notifications, assets and audit events.
- Return a stable error code, human-safe message, request_id and optional field errors.
- Use ETag/version fields or If-Match for edits that can conflict with approval.
- Use job resources for long-running generation/rendering: return 202 with job ID and poll or stream status.
- Never accept workspace_id from client without server-side membership validation.
- Rate-limit public auth, approval deep links and webhook endpoints separately.

15.2 Example error envelope

```
{
  "error": {
    "code": "POST_VERSION_CONFLICT",
    "message": "This post changed after the approval request was sent.",
    "action": "REFRESH_AND_REVIEW",
    "request_id": "req_...",
    "details": {"expected_version": 7, "current_version": 8}
  }
}
```

```
}  
}
```

16

Security, privacy and compliance

Social tokens, approval authority and customer brand data are high-value assets.

16.1 Security controls

Control	Implementation
Token encryption	KMS envelope encryption; separate encrypted credentials from normal connection metadata.
Secret access	Workers receive scoped decryption permissions; browser never receives provider access tokens.
OAuth protection	State, nonce/PKCE where supported, exact redirect allowlist, short-lived callback state and replay prevention.
Webhook verification	Verify provider signature against raw request body; rate limit and deduplicate before processing.
RBAC and tenancy	Server-side membership/role guards and workspace-scoped repositories; automated cross-tenant tests.
Audit logging	Immutable log for connection, approval, schedule, publish, billing, admin and data-deletion actions.
Sensitive actions	Recent-auth requirement, OTP or confirmation for Autopilot, disconnect, member removal and data deletion.
Media safety	Virus/malware scanning, MIME sniffing, EXIF stripping, size limits and signed uploads.
Data minimization	Store only required WhatsApp phone and consent evidence; redact third-party payloads and logs.
Backups	Encrypted automated DB backups, point-in-time recovery and tested restoration runbook.
Incident response	Provider credential revocation procedure, customer notification workflow and security event monitoring.

16.2 Privacy and consent

- Capture separate consent for social-account connection, WhatsApp notifications, analytics and optional AI processing disclosures.
- Provide account export and deletion workflows; define retention for drafts, published records, logs and provider payloads.
- Keep advertiser/customer conversations private from unrelated tenants and staff; support audited support access only.
- Prepare for India's data-protection obligations with purpose limitation, consent records, grievance/contact process and processor/vendor inventory. Obtain legal review before launch.
- Never use uploaded client assets to train a shared model unless separately and explicitly agreed.

16.3 Content safety and regulated categories

Risk	MVP policy
Medical/health claims	Always approval-required; use verified source facts and mandatory disclaimer rules.
Financial/legal claims	Always approval-required; no inferred returns, guarantees or legal advice.
Religious/festival content	Use user-selected festivals and region; sensitive templates require explicit approval.
Prices/offers	Only from structured active OfferReference supplied/confirmed by client.
Personal images/testimonials	Require rights/consent status on asset; block unknown rights from automatic use.
Political content	Exclude from automatic generation in MVP.

17

Observability, QA and operational tooling

External APIs fail; the product must explain, reconcile and recover.

17.1 Observability

Signal	Required dimensions / alerts
Structured logs	request_id, workspace_id, brand_id, job_id, provider, entity ID, attempt, error code; never raw tokens.
Tracing	Browser → API → outbox → queue → worker → provider; propagate correlation IDs.
Metrics	Generation success, schema repair rate, render latency, approval time, publish success, retries, webhook lag, queue depth.
Business metrics	Onboarding completion, account connection, first approval, first publish, monthly active brands, Autopilot adoption, churn.
Cost metrics	AI cost/post, render/storage, WhatsApp delivery, support time and provider call volume.
Alerts	High failure rate per provider, token refresh spike, queue backlog, webhook signature failures, publishing deadline misses.

17.2 Test strategy

Test layer	Coverage
Unit	State transitions, approval policies, slot builder, validators, retry classifier, entitlement logic.
Schema/AI	Golden inputs, strict JSON parsing, forbidden-claim tests, multilingual output, prompt regression and cost threshold.
Renderer	Snapshot/golden images at every ratio, overflow, font fallback, RTL readiness, low-resolution assets.
Integration	PostgreSQL transaction/outbox, Redis queue, storage and provider adapter mocks.
Contract	Recorded/sanitized provider fixtures and versioned response parsers.
Webhook	Signature, duplicate delivery, reordering, delayed events, malformed payload and one-time approval token.
End-to-end	Signup → onboarding → plan → approval → schedule → mocked publish → report on desktop and mobile.
Reliability	Worker crash mid-call, DB unavailable, queue retry, provider timeout and ambiguous publish reconciliation.
Security	Cross-tenant access, IDOR, OAuth state replay, webhook forgery, signed-link expiry and privilege escalation.
Accessibility	Keyboard navigation, focus states, labels, contrast, screen-reader status and mobile tap targets.

17.3 Internal operations console

- Provider health dashboard by platform and app version.
- Failed publish queue with normalized error, raw-reference access and safe retry/reconcile actions.
- Webhook inbox with signature status, dedupe status and replay control.
- Connection diagnostic without exposing token value.
- Prompt and model release control with canary percentage and rollback.
- Template version release, preview and rollback.
- Festival catalogue review and date/version publishing.
- Customer support impersonation only through audited, time-limited access with visible banner.

18

Delivery sequence and acceptance criteria

Build vertical slices that publish successfully before expanding feature breadth.

18.1 Recommended implementation sequence

Sequence	Deliverable
Foundation	Workspace/brand tenancy, auth/RBAC, database, storage, queues, outbox, audit log, design tokens and CI/CD.
Onboarding vertical slice	Brand setup, assets, preferences, social connection shell, WhatsApp opt-in and first sample plan.
Content engine	Slot builder, AI schemas/model router, Brand Brain retrieval, template renderer and post editor.
Approval slice	In-app approval, versions, batch approval, notification records and WhatsApp Cloud API webhook loop.
Publishing slice	Meta Instagram/Facebook provider adapters, schedule/preflight/publish jobs, status, retries and confirmation.
GBP slice	Google OAuth/location selection, Local Posts adapter, per-platform variant and insights job.
Commercial slice	Billing/entitlements, usage, reports, support/admin tooling and production observability.
Phase 2	LinkedIn, agency dashboard, multi-location, regional-language improvements, advanced analytics and optional Temporal migration.

18.2 MVP acceptance criteria

- ☐ A new user can create a workspace and brand on desktop or mobile and resume interrupted onboarding.
- ☐ The user can upload a logo and images, set voice/rules, select relevant festivals and connect at least one launch platform.
- ☐ Creatyv can generate a valid seven-day or monthly plan using deterministic slots and schema-valid AI output.
- ☐ At least one template creative renders in 1:1 and 4:5 without overflow and can be edited through constrained controls.
- ☐ A post can move through review, WhatsApp or in-app approval, scheduling and per-platform publishing with a complete audit trail.
- ☐ A WhatsApp Approve button validates the sender/request/version and causes exactly one approval transition and exactly one schedule set.
- ☐ Instagram/Facebook and GBP publishing use direct official APIs; no Zapier/Make step is required.
- ☐ Failed or partially published posts are visible, classified and retryable per destination.
- ☐ Connection expiry/revocation produces a user-facing reconnect state and prevents unsafe publishing.
- ☐ All provider tokens are encrypted, webhooks verified and cross-tenant authorization tested.
- ☐ Home and mobile views prioritize pending approvals, today's posts, failures and next required action.
- ☐ Basic monthly activity and available performance metrics are visible without blocking on every provider metric.
- ☐ Internal ops can inspect, reconcile and retry failures without direct database edits.

18.3 Launch gates

Gate	Required evidence
Product	At least 10 pilot brands complete onboarding and publish through the system.
Reliability	Successful publish rate and retry behavior meet internal target for each launch provider.
Content quality	Pilot approval rate, edit distance and restricted-claim incidents are reviewed by vertical.

Gate	Required evidence
Security	External or independent security review of auth, OAuth, webhooks, tenant isolation and token storage.
Platform compliance	Meta and Google app access/review complete; policies, deletion URL and consent flows approved.
Operations	On-call ownership, provider incident playbook, customer support flow and rollback procedure documented.
Commercial	Plan entitlements, limits, billing webhooks, cancellation and refund policy reflected in product.

Recommended pilot

Use MMM-managed hospitality, restaurant, retail and service brands as controlled pilots. Begin in approval-required mode, compare generated versus edited output, and unlock Autopilot only after a brand has a stable acceptance history.

Official reference links and implementation checklist

These are the starting points; verify current versions, scopes and review requirements during engineering.

OpenAI model catalog: <https://developers.openai.com/api/docs/models>

OpenAI Structured Outputs: <https://developers.openai.com/api/docs/guides/structured-outputs>

OpenAI image generation: <https://developers.openai.com/api/docs/guides/image-generation>

OpenAI embeddings: <https://developers.openai.com/api/docs/guides/embeddings>

OpenAI moderation: <https://developers.openai.com/api/docs/guides/moderation>

Instagram Content Publishing: <https://developers.facebook.com/documentation/instagram-platform/content-publishing>

IG User Media container: <https://developers.facebook.com/documentation/instagram-platform/instagram-graph-api/reference/ig-user/media>

IG User Media Publish: https://developers.facebook.com/documentation/instagram-platform/instagram-graph-api/reference/ig-user/media_publish

WhatsApp Business Platform overview: <https://developers.facebook.com/documentation/business-messaging/whatsapp/about-the-platform>

WhatsApp interactive reply buttons: <https://developers.facebook.com/documentation/business-messaging/whatsapp/messages/interactive-reply-buttons-messages>

WhatsApp messages webhook: <https://developers.facebook.com/documentation/business-messaging/whatsapp/webhooks/reference/messages>

Google Business Profile APIs: <https://developers.google.com/my-business>

Create Google Business Profile posts: <https://developers.google.com/my-business/content/posts-data>

Google Local Posts create reference: <https://developers.google.com/my-business/reference/rest/v4/accounts.locations.localPosts/create>

LinkedIn API access: <https://learn.microsoft.com/en-us/linkedin/shared/authentication/getting-access>

LinkedIn organization access control: <https://learn.microsoft.com/en-us/linkedin/marketing/community-management/organizations/organization-access-control-by-role?view=li-lms-2026-06>

Reference review date: 17 July 2026. Third-party documentation, API versions, quotas, permissions and pricing can change. Engineering must pin versions where possible, subscribe to provider changelogs, and test changes in staging before production rollout.

19.1 Pre-development decisions to lock

- ☐ Choose AWS S3/CloudFront versus Cloudflare R2/CDN and document provider-fetchable media URL behavior.
- ☐ Choose managed identity provider and OTP vendor; define account recovery and phone/email ownership rules.
- ☐ Create Meta and Google developer projects immediately and start business verification/app access work.
- ☐ Confirm launch platforms and post formats: Instagram feed/carousel/reel cover, Facebook Page posts and GBP update/event/offer.
- ☐ Approve initial template system, aspect ratios, licensed fonts and five to ten vertical templates.
- ☐ Define regulated vertical policy and which brands require mandatory human approval.
- ☐ Finalize plan entitlements, post limits, brands, platforms and AI/render usage limits.
- ☐ Create pilot brands and golden content sets for hospitality, restaurants, retail and local services.
- ☐ Approve data retention, WhatsApp consent, privacy policy, terms, cancellation and support policies with counsel.
- ☐ Assign owners for provider integrations, AI quality, template rendering, security and operations.

Final build recommendation

The smallest commercially strong Creatyv is an approval-to-publish operating system.

Recommended build

Ship a responsive PWA with Brand Brain onboarding, deterministic monthly planning, template creatives, WhatsApp/in-app approvals, direct Meta and GBP publishing, reliable retries and a transparent action-first dashboard. Keep AI, providers and templates replaceable behind versioned interfaces.

The architecture should optimize for trust and recoverability before feature count. A client must always know what is being created, what needs approval, where and when it will publish, and what happened if a provider failed. When that loop works repeatedly, Creatyv becomes sellable as Social Autopilot; advanced analytics, LinkedIn, agency workspaces, multi-location and richer generative media can then be added without rewriting the core.

END OF SPECIFICATION

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